

A
Project Report
ON
“Placement Management System”

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Master of Computer Application
Savitribai Phule Pune University

Vidya Pratishthan’s Institute of Information Technology (VIIT),
(ACCREDITED BY NAAC WITH “B” GRADE, CYCLE-II)

Vidya Nagari, Baramati,
Dist. Pune 413 133
(2025-26)



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CERTIFICATE

This is to certify that, Mr. **Pradeep Ramesh Taware** of MCA Course, SEM- II have completed their Project Work titled ' **Placement Management System** ', as a part of the curriculum, during the Academic Year 2025-26.

University Seat No:

Dr. Ajit Adsul

Head of Department

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External Examiner

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2. _____

Sign of Examiners:

Date of Examination:



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GUIDE CERTIFICATE

This is to certify that Mr **Pradeep Ramesh Taware** of the MCA Course, SEM- II have completed **their** Project Work titled “**Placement Management System**”, under my guidance during the Academic Year 2025-26.

Mr. Pravin Bagal
Project Guide

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1. Introduction

1.1 Introduction to the System :

The **Placement Management System** is a web-based or software application developed to automate and manage the entire campus recruitment process efficiently. It acts as a centralised platform that connects students, placement officers, and recruiting companies, reducing manual work and improving communication. In traditional placement processes, maintaining student records, company details, job notifications, and recruitment activities manually becomes time-consuming and error-prone. The Placement Management System solves this problem by digitising all placement-related operations such as student registration, eligibility verification, company postings, interview scheduling, and result management. The system allows students to create profiles, upload resumes, and apply for job opportunities based on eligibility criteria. Placement officers can manage student data, track recruitment drives, and monitor placement statistics, while companies can post job requirements and shortlist candidates easily.

1.2 Existing System

In the existing placement process, most of the activities are carried out manually using paperwork, spreadsheets, emails, or notice boards. Student information, academic records, and placement details are maintained separately, which makes data management difficult and time-consuming. Students usually receive placement updates through manual announcements or notices, which may lead to delays or missed opportunities. The placement officer needs to verify student eligibility, manage company details, and schedule interviews manually, increasing workload and chances of human error. Additionally, maintaining records of placed and unplaced students becomes complicated due to the absence of a centralised system. Generating placement reports and tracking recruitment progress also requires significant manual effort.

1.3 Proposed System

The proposed **Placement Management System** is an automated, centralised software solution designed to streamline and manage all placement-related activities. The system aims to replace the manual placement process with a digital platform that ensures faster communication, accurate data handling, and improved coordination between students, placement officers, and recruiting companies. In the proposed system, students can register online, create their profiles, upload resumes, and apply for job opportunities based on eligibility criteria. Placement officers can manage student records, company information, job postings, interview schedules, and placement results through a single platform. Companies can also access the system to post job requirements and shortlist eligible candidates. The system maintains all placement data in a centralised database,

reducing paperwork and minimising human error. Automated notifications help students stay updated about recruitment drives, interview schedules, and results.

1.4 Scope of System

The **Placement Management System** is designed to manage and automate the complete placement process within an educational institution. The scope of the system covers all activities related to student registration, company recruitment drives, and placement record management. The system provides a common platform for students, placement officers, and recruiters to interact efficiently. Students can register, maintain their academic profiles, upload resumes, and apply for available job opportunities. Placement officers can manage student information, company details, job notifications, eligibility criteria, and interview schedules. Recruiters can view eligible candidates and conduct recruitment processes smoothly. The system also helps institutions maintain accurate placement records and generate reports for analysis and future reference.

1.5 Operating Environment

1.5.1 Hardware Requirement

1. **Processor:** Intel Core i3 or higher
2. **RAM:** 4 GB or higher
3. **Hard Disk:** Minimum 500 GB free space

1.5.2 Software Requirement

1. **Operating System:** Windows 10 / 11 or Linux
2. **Front-End Tool:** HTML, CSS, Bootstrap, React JS
3. **Back-End:** Java
4. **Database:**MySQL

2. Analysis and Design

1.1 Fact-Finding Techniques

Fact-finding techniques are methods used to collect information and understand the requirements of a system before developing it. These techniques help in identifying problems in the existing system and gathering accurate data needed for designing the **Placement Management System**. During system analysis, different fact-finding methods are used to study current placement activities, user requirements, and operational challenges faced by students and placement officers.

1.1.1 Interview

An **Interview** is one of the most effective fact-finding techniques used during system analysis to collect detailed information directly from users and stakeholders. It involves face-to-face or online interaction between the system analyst and the individuals involved in the existing system. In the development of the **Placement Management System**, interviews were conducted with placement officers, students, and administrative staff to understand the current placement procedures, problems faced, and expectations from the new system.

1. Structured Interview :

A **Structured Interview** is a type of interview in which a set of predefined and organized questions is prepared before conducting the interview. All participants are asked the same questions in the same order to collect specific and consistent information.

2. Unstructured Interview :

An **Unstructured Interview** is an informal interview method where questions are not fixed in advance. The discussion flows naturally, allowing respondents to freely express their opinions, experiences, and suggestions.

1.1.2 Observation

Observation is an important fact-finding technique used in system analysis to understand how the existing system operates in real-time. In this method, the system analyst carefully watches the activities, processes, and workflow performed by users without directly interfering in their work.

2.2 Feasibility of the System

2.2.1 Technical feasibility

Technical feasibility refers to the evaluation of whether the required technology, hardware, software, and technical resources are available to develop and implement the proposed Placement Management System successfully. The proposed system can be developed using commonly available technologies such as programming languages, databases, and web development tools. The required hardware and software resources are easily accessible within educational institutions, making the system technically feasible. The system does not require advanced or expensive infrastructure and can operate efficiently on standard computers with internet connectivity. Existing technical knowledge of developers and administrators is sufficient to design, implement, and maintain the system.

2.2.2 Economical feasibility

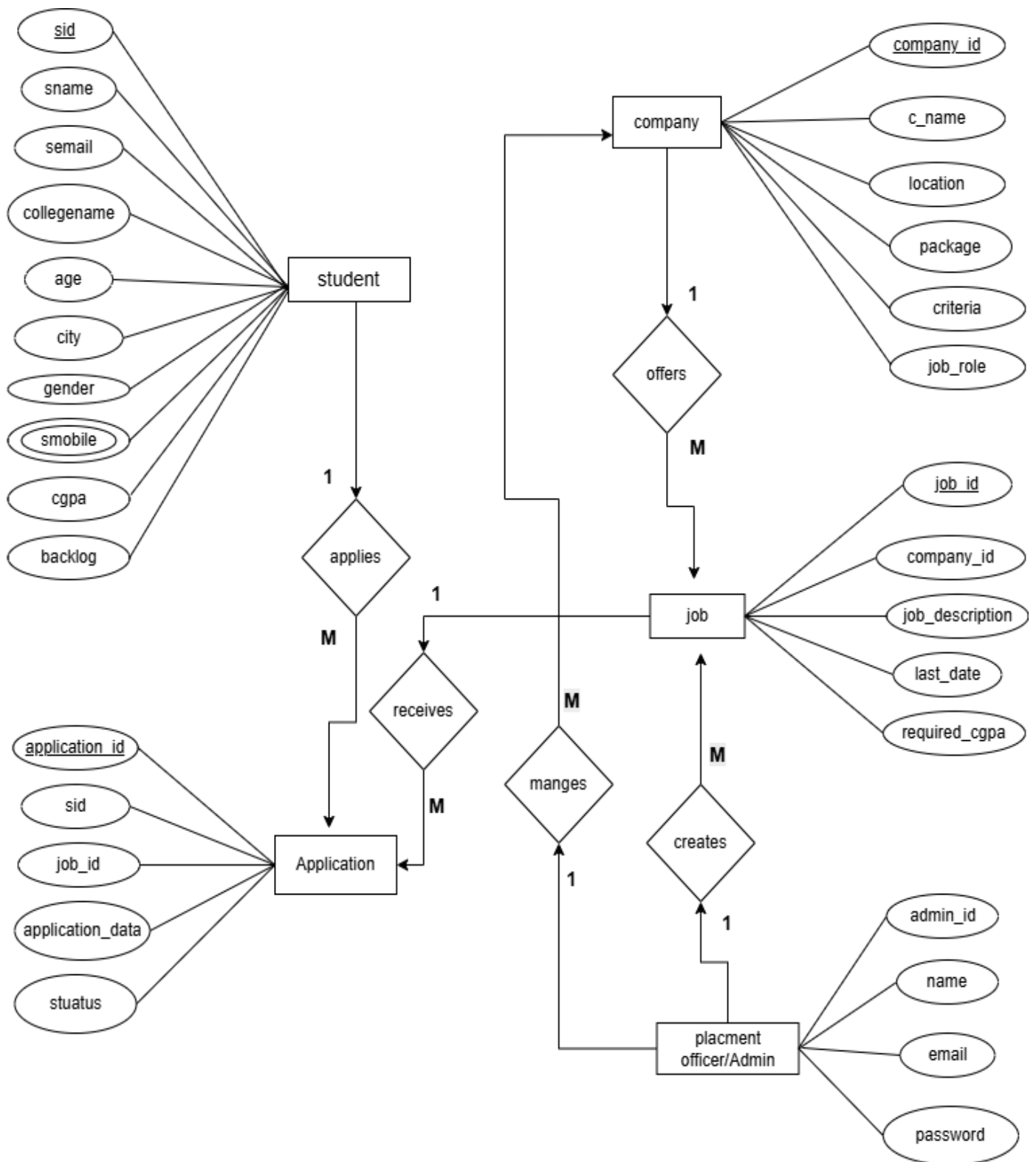
Economic feasibility evaluates the cost effectiveness of the proposed **Placement Management System** by comparing the development and operational costs with the expected benefits gained from the system. The proposed system is economical because it minimizes manual work, reduces paperwork, and saves administrative time and resources. Since the system can be developed using existing hardware and commonly available software tools, the overall implementation cost remains low.

2.2.3 Operational feasibility

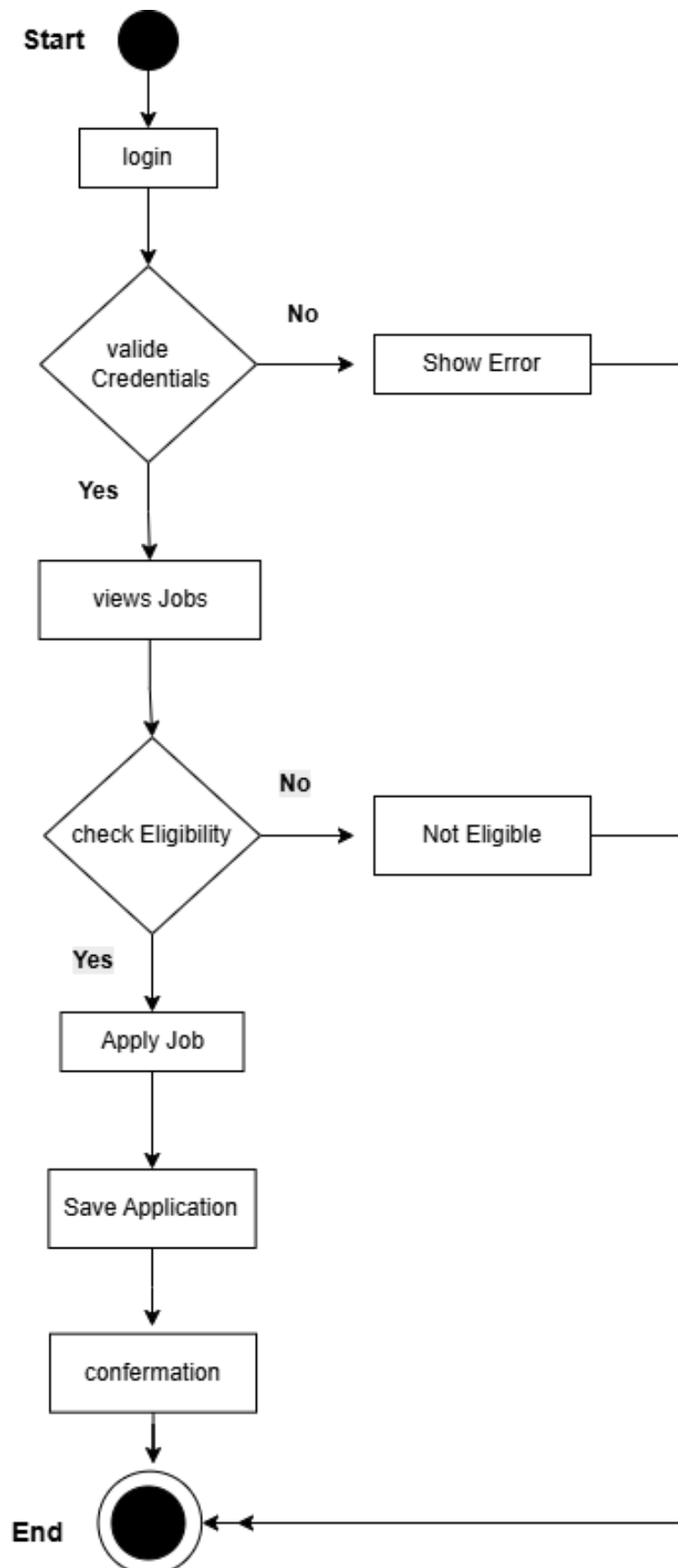
Operational feasibility refers to the ability of the proposed Placement Management System to function effectively within the existing organizational environment and meet user requirements. It evaluates whether the system will be accepted and efficiently used by students, placement officers, and administrative staff. The proposed system is user-friendly and designed to simplify placement-related activities such as student registration, job notifications, eligibility checking, and report management. Since the system automates routine tasks, it reduces manual workload and improves overall operational efficiency.

2.3 Functional & Non-Functional Requirements Diagrams:

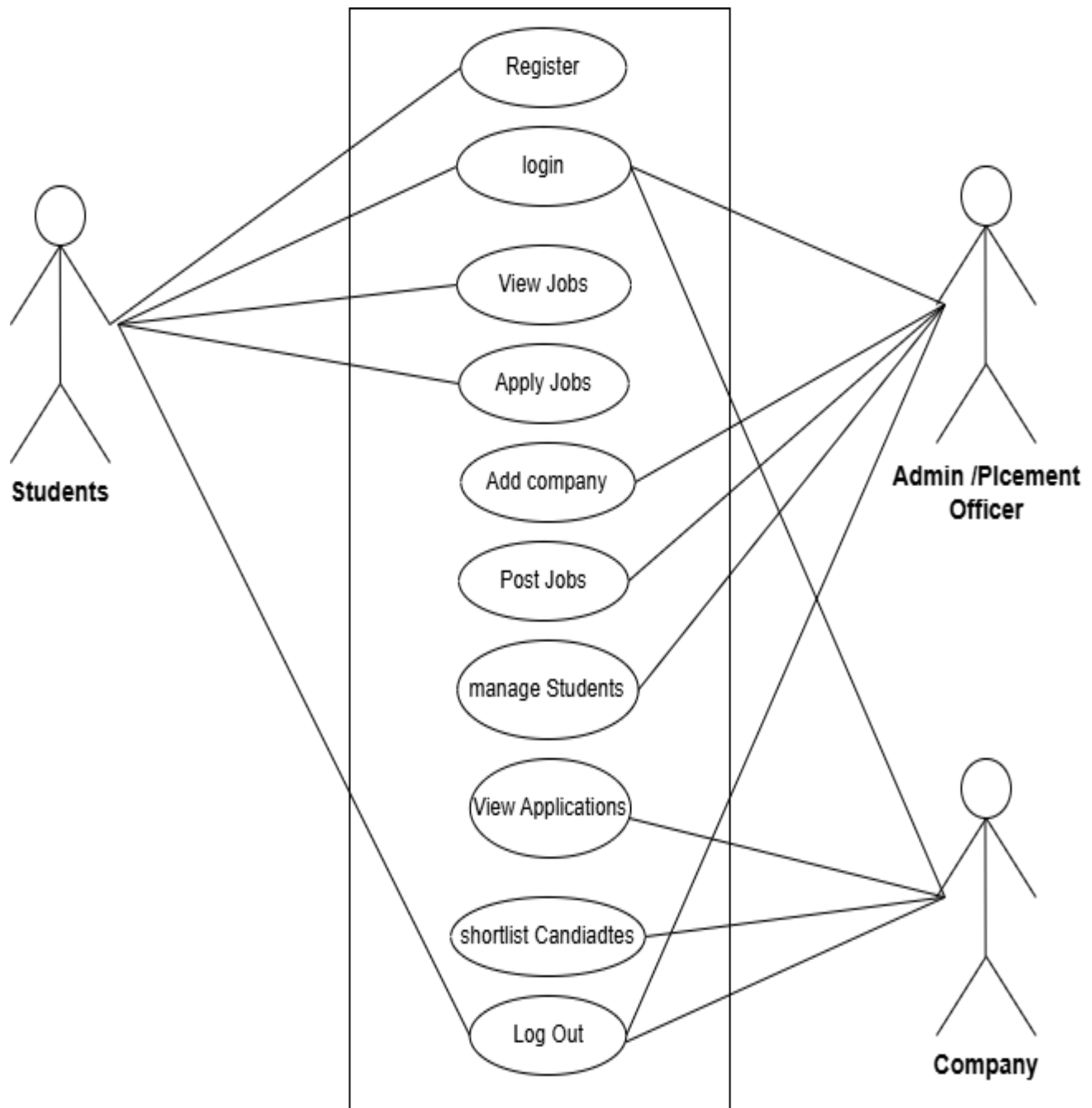
2.3.1 ER Diagram



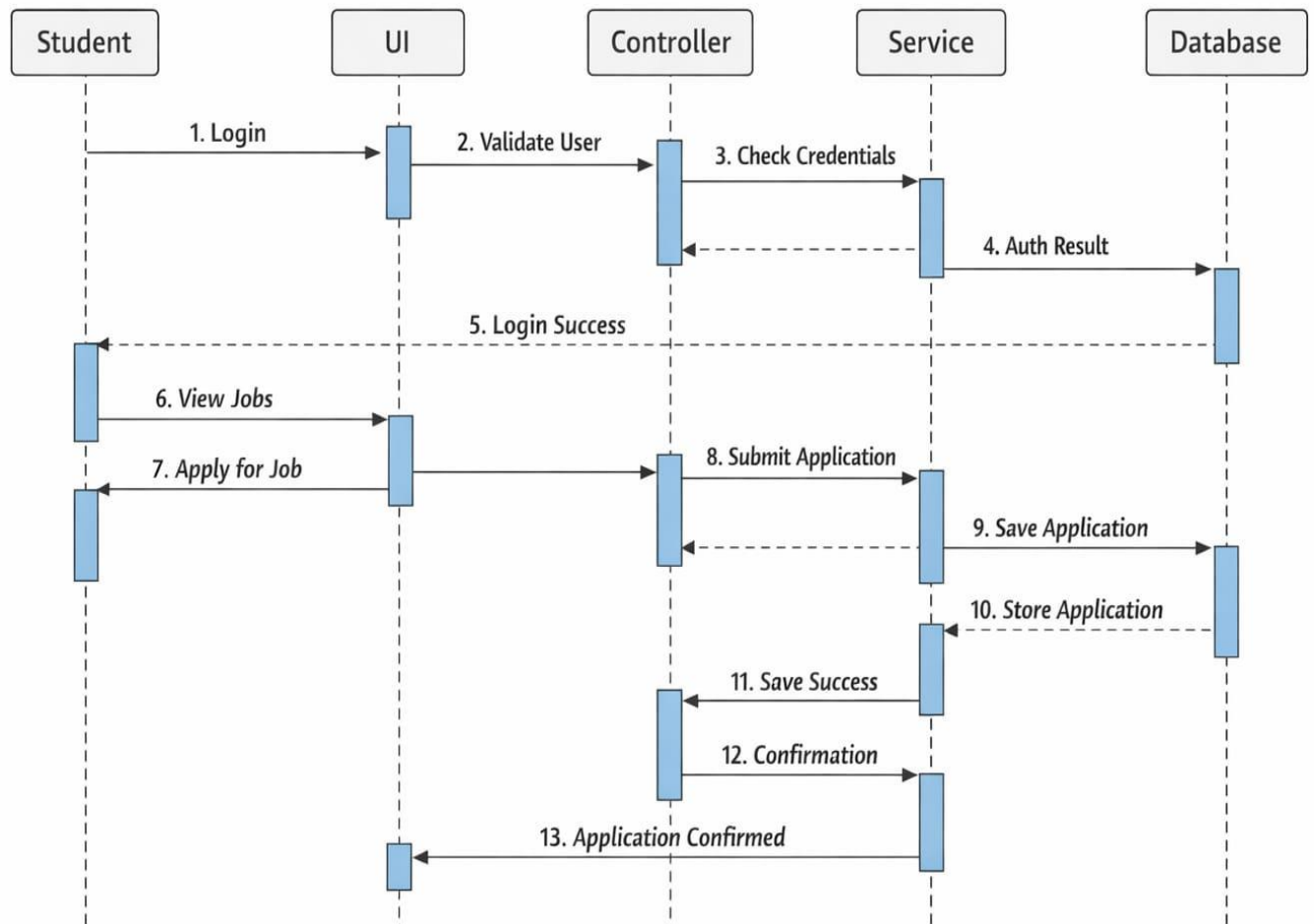
2.3.2 Activity Diagram



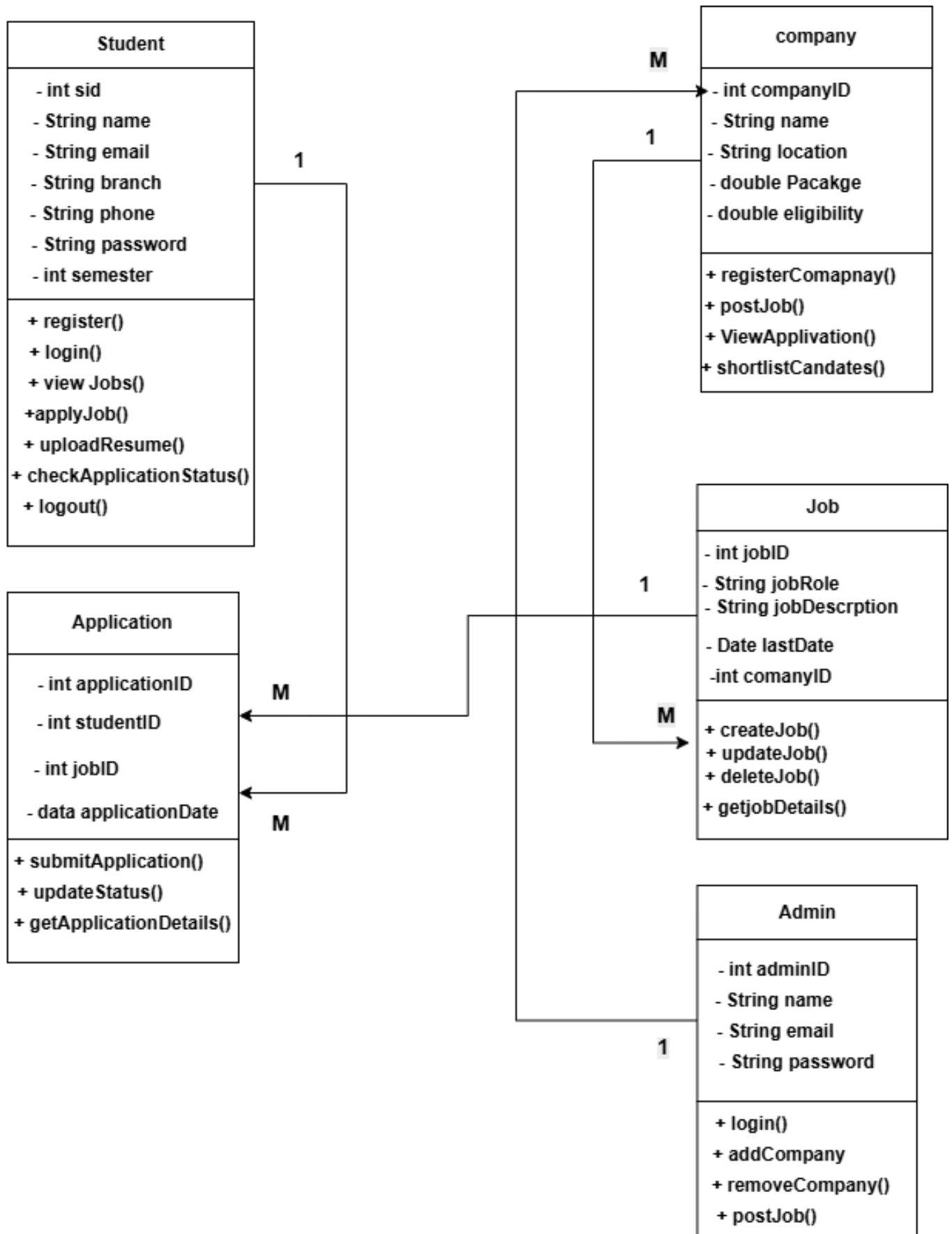
2.3.3 Use case Diagram



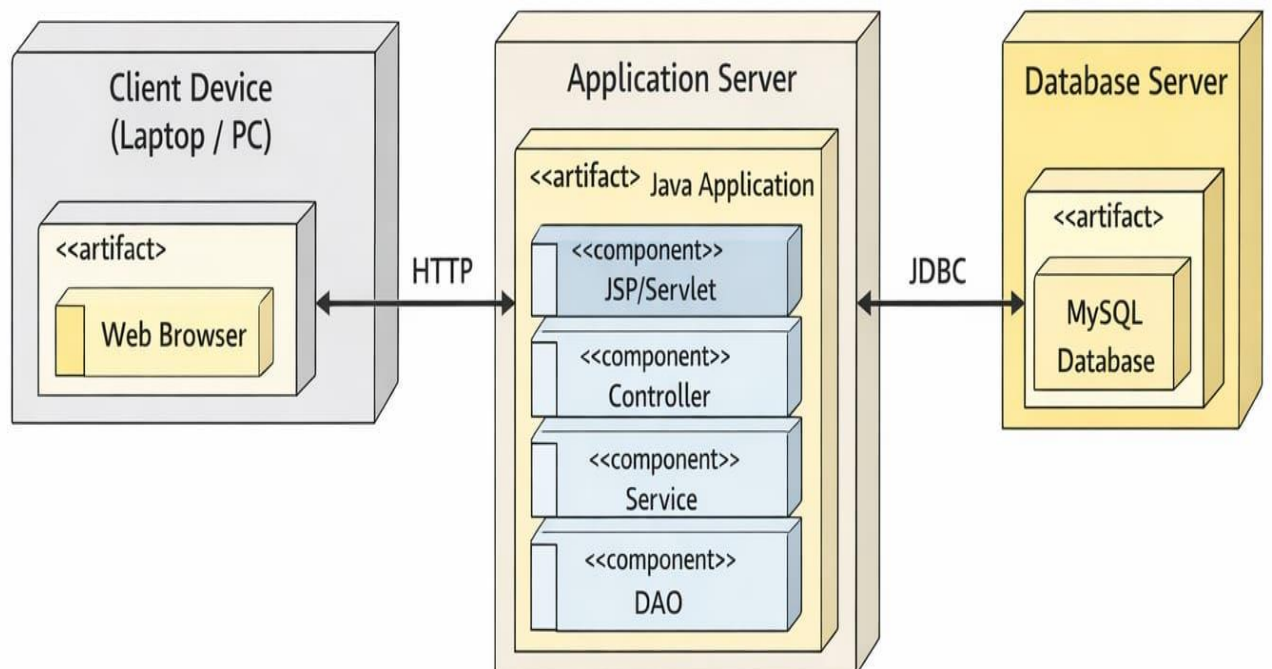
2.3.4 Sequence Diagram



2.3.5 Class Diagram

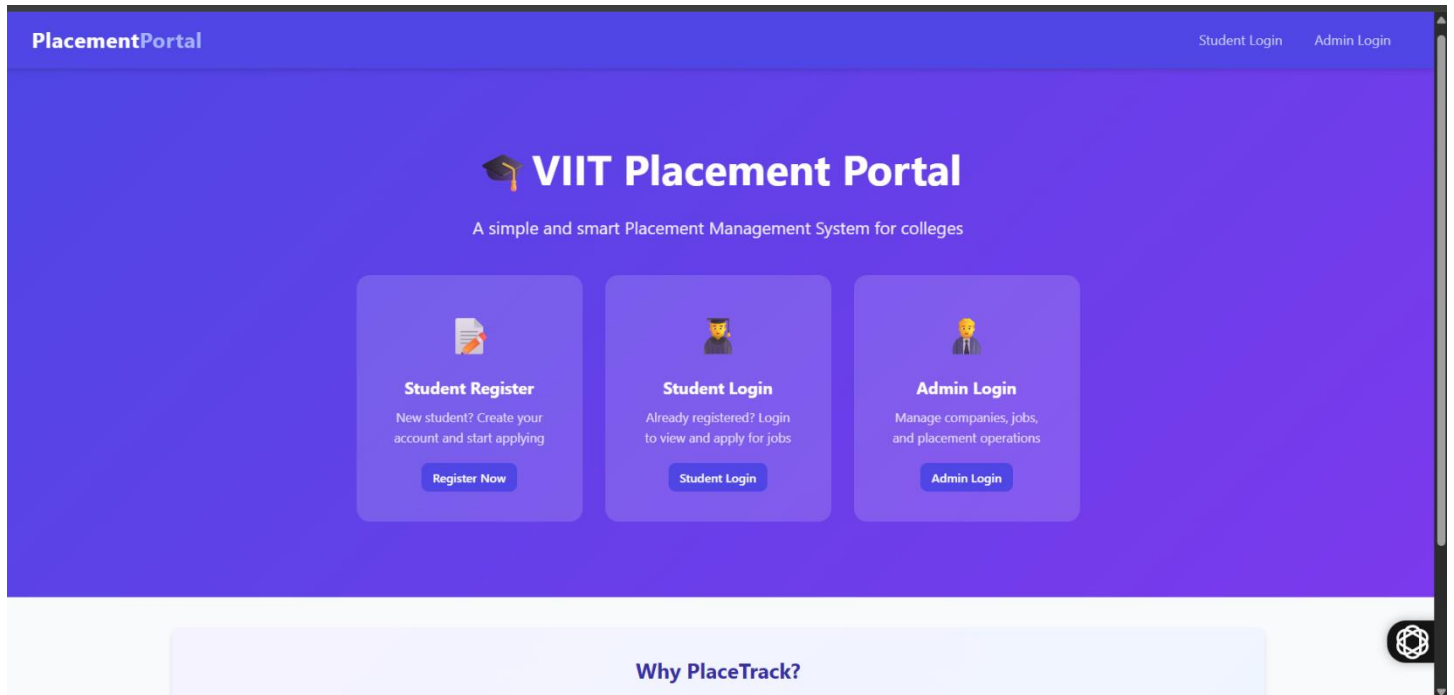


2.3.6 Deployment Diagram

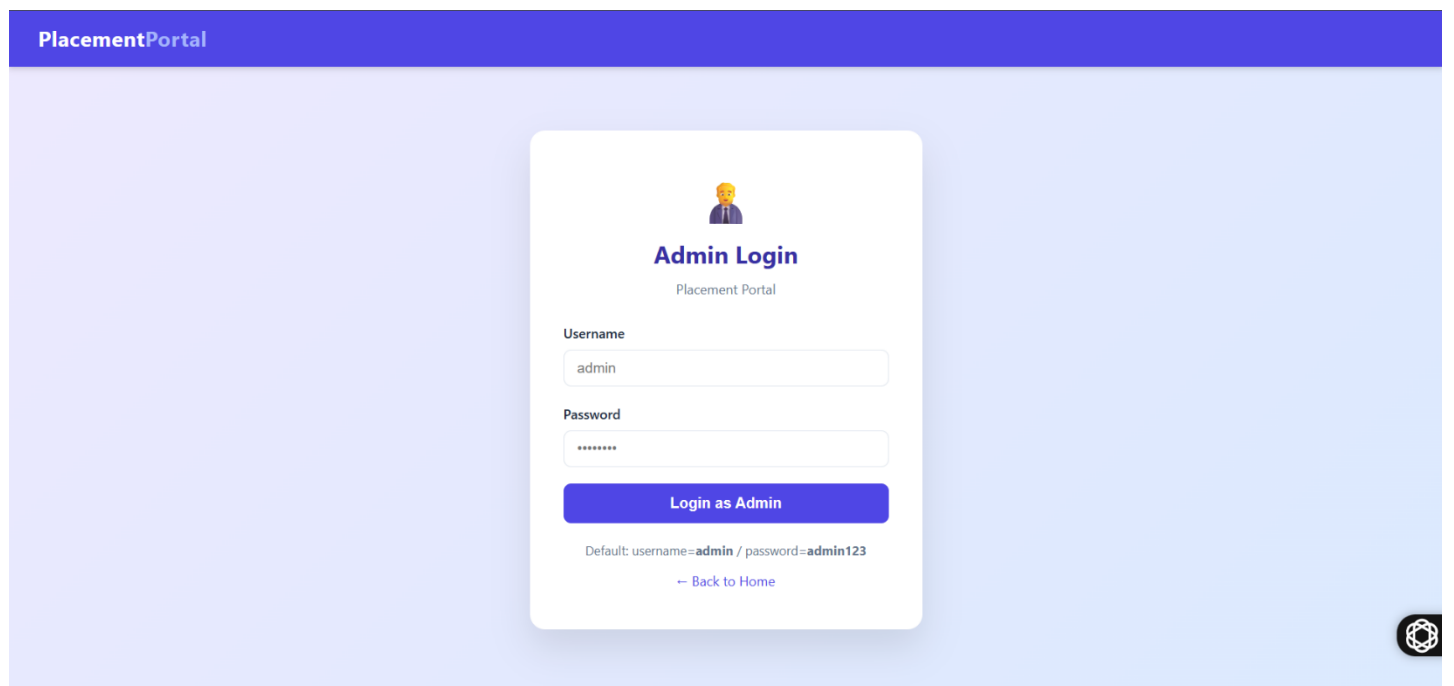


3. Design

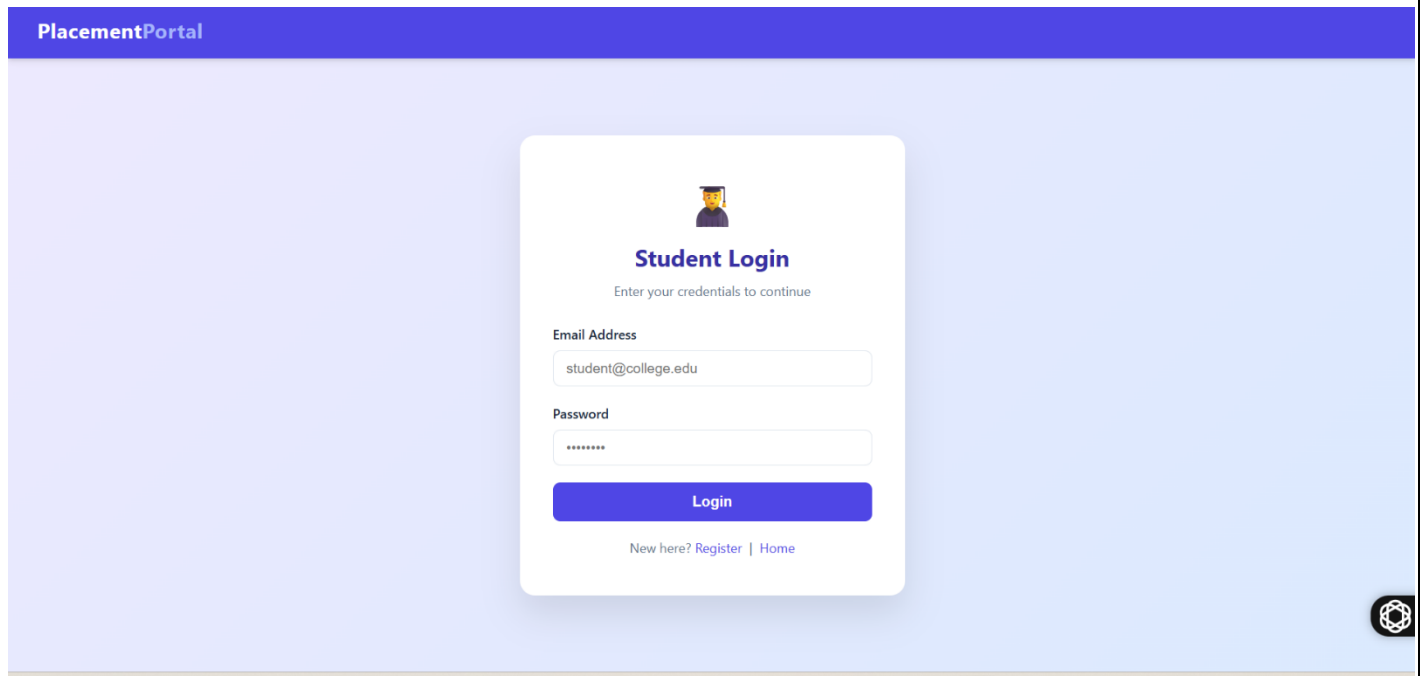
1. Landing Page



2. Admin Login :

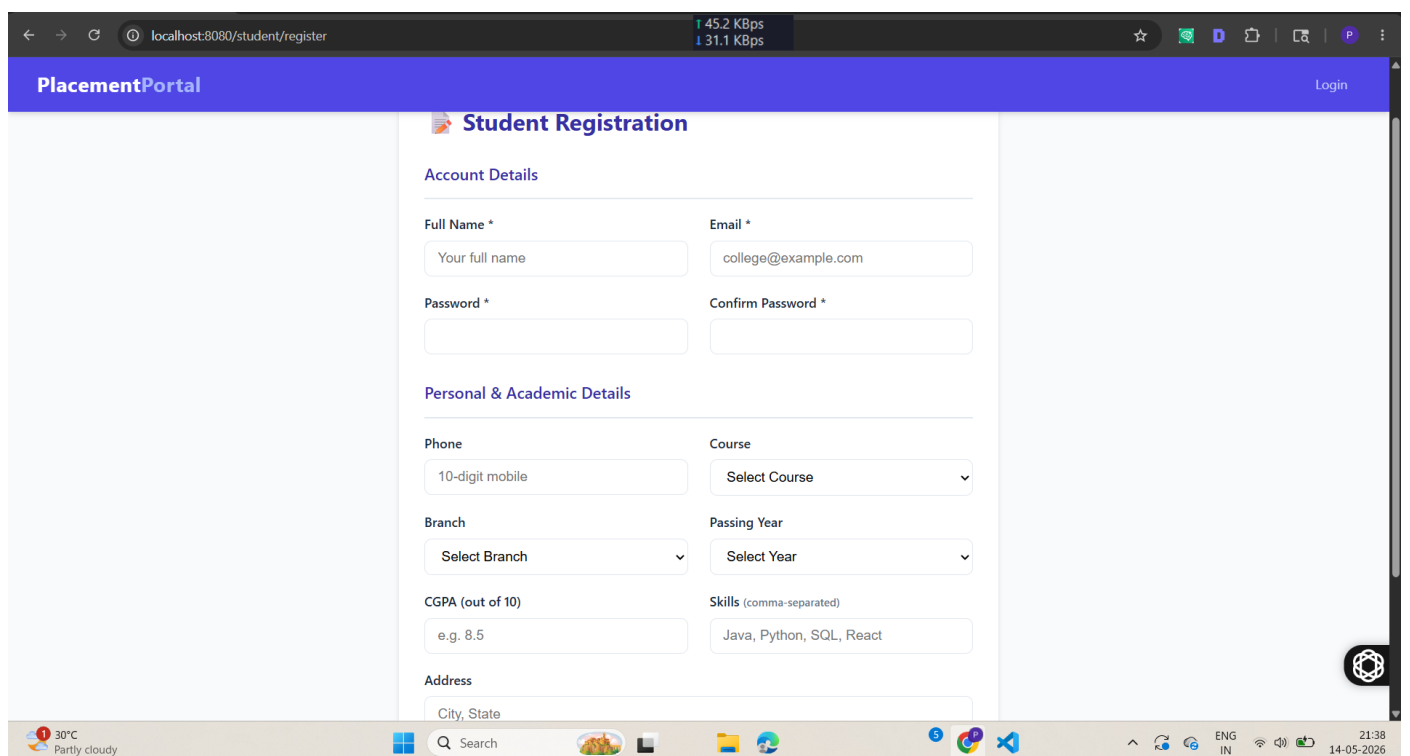


3. Student Login :



The screenshot shows the 'Student Login' page of the 'PlacementPortal'. The page has a blue header with the text 'PlacementPortal'. The main content area is light blue. In the center, there is a white card with a blue header 'Student Login' and a subtitle 'Enter your credentials to continue'. Below this, there are two input fields: 'Email Address' with the value 'student@college.edu' and 'Password' with masked characters '*****'. A blue 'Login' button is positioned below the password field. At the bottom of the card, there is a link 'New here? Register | Home'. A small circular icon is visible in the bottom right corner of the page.

4. Student Registration :



The screenshot shows the 'Student Registration' page of the 'PlacementPortal'. The page has a blue header with the text 'PlacementPortal' and a 'Login' link on the right. The main content area is light blue. The registration form is titled 'Student Registration' and is divided into two sections: 'Account Details' and 'Personal & Academic Details'. The 'Account Details' section includes fields for 'Full Name *', 'Email *', 'Password *', and 'Confirm Password *'. The 'Personal & Academic Details' section includes fields for 'Phone', 'Course', 'Branch', 'Passing Year', 'CGPA (out of 10)', 'Skills (comma-separated)', and 'Address'. The 'Phone' field has a placeholder '10-digit mobile'. The 'Course' field is a dropdown menu with 'Select Course' as the selected option. The 'Branch' field is a dropdown menu with 'Select Branch' as the selected option. The 'Passing Year' field is a dropdown menu with 'Select Year' as the selected option. The 'CGPA (out of 10)' field has a placeholder 'e.g. 8.5'. The 'Skills (comma-separated)' field has a placeholder 'Java, Python, SQL, React'. The 'Address' field has a placeholder 'City, State'. A small circular icon is visible in the bottom right corner of the page.

5. Student Dashboard :

The screenshot shows the Student Dashboard of the PlacementPortal. The header is blue with the text "PlacementPortal" and navigation links: "Dashboard", "Browse Jobs", "Profile", and "Logout". The main content area has a light blue background. At the top, it says "Welcome, Pradeep Taware" with a hand icon. Below this are three white cards: "0 Applications" with a clipboard icon, "8.9 CGPA" with a graduation cap icon, and "NOT_PLACED Status" with a trophy icon. A "My Profile" section follows, containing fields for Email, Course, Branch, Passing Year, Phone, CGPA, Skills, and Address, along with "Edit Profile" and "Browse Jobs" buttons. Below that is a "My Applications" section with a message: "You haven't applied to any jobs yet. Browse Jobs →". The bottom of the image shows a Windows taskbar with the date 14-05-2026 and time 21:39.

PlacementPortal

Dashboard Browse Jobs Profile Logout

Welcome, Pradeep Taware 🙌

0 Applications

8.9 CGPA

NOT_PLACED Status

My Profile

Email: pradeeptaware081@gmail.com
Course: MBA
Branch: CE
Passing Year: 2024
Phone: 8421077165
CGPA: 8.9
Skills: Java
Address: मालेगाव बुद्रुक, ता- बारामती, जि - पुणे 413115

Edit Profile Browse Jobs

My Applications

You haven't applied to any jobs yet. Browse Jobs →

30°C Partly cloudy

Search

ENG IN

21:39 14-05-2026

6. Student Edit Profile :

The screenshot shows the "Edit Profile" page of the PlacementPortal. The header is blue with the text "PlacementPortal" and navigation links: "Dashboard", "Browse Jobs", "Profile", and "Logout". The main content area has a light blue background. The "Edit Profile" section is centered and contains two sub-sections: "Personal Information" and "Academic Information". The "Personal Information" section has fields for Full Name, Phone, and Address. The "Academic Information" section has fields for Course, Branch, Passing Year, CGPA (out of 10), and Skills (comma-separated). The "Skills" field contains the text "Java".

PlacementPortal

Dashboard Browse Jobs Profile Logout

Edit Profile

Personal Information

Full Name Phone
Pradeep Taware 8421077165

Address
मालेगाव बुद्रुक, ता- बारामती, जि - पुणे 413115

Academic Information

Course Branch
MBA CE

Passing Year CGPA (out of 10)
2024 8.9

Skills (comma-separated)
Java

7. Admin Dashboard :

The Admin Dashboard in the PlacementPortal features a top navigation bar with links to Dashboard, Companies, Jobs, Applications, Students, Placed, and Logout. The main content area is titled "Admin Dashboard" and displays five key metrics: 503 Total Students, 53 Companies, 18 Job Postings, 21 Applications, and 250 Placed Students. Below these metrics is a "Quick Actions" section with buttons for Add Company, Post Job, View Applications, Placed Students, and All Students. A small globe icon is visible in the bottom right corner.

PlacementPortal Dashboard Companies Jobs Applications Students Placed Logout

Admin Dashboard

503 Total Students

53 Companies

18 Job Postings

21 Applications

250 Placed Students

Quick Actions

- + Add Company
- Post Job
- View Applications
- Placed Students
- All Students

8. Available Jobs :

The Available Job Openings page in the PlacementPortal shows a top navigation bar with links to Dashboard, Browse Jobs, Profile, and Logout. The main content area is titled "Available Job Openings" and displays six job listings. Each listing includes the job title, company, location, package, minimum CGPA, branch, passing year, deadline, and skills. A red "Not Eligible" message is shown for each job.

PlacementPortal Dashboard Browse Jobs Profile Logout

Available Job Openings

software Developer
TCS | Pune
Package: 4.5 LPA
Min CGPA: 6.5
Branch: IT
Passing Year: 2025
Deadline: 08 May 2026
Skills: Java
Not Eligible

Software Tester
Cognizant | Chennai
Package: 3.5 LPA
Min CGPA: 7.0
Branch: IT
Passing Year: 2024
Deadline: 10 Jun 2026
Skills: Selenium
Not Eligible

Support
Software Tech | Baramati
Package: 6.0 LPA
Min CGPA: 6.0
Branch: IT
Passing Year: 2024
Deadline: 08 May 2026
Skills: SQL, Java
Not Eligible

Software Engineer
TCS | Pune
Package: 4.5 LPA
Min CGPA: 6.5
Branch: Computer Science
Passing Year: 2025
Deadline: 20 May 2026
Skills: Java, Spring Boot, MySQL

Associate Engineer
Software Tech | Baramati
Package: 4.2 LPA
Min CGPA: 6.0
Branch: Information Technology
Passing Year: 2025
Deadline: 22 May 2026
Skills: Java, SQL, OOP

Full Stack Developer
Cognizant | Chennai
Package: 4.8 LPA
Min CGPA: 6.2
Branch: Computer Science
Passing Year: 2025
Deadline: 24 May 2026
Skills: JavaScript, React, Node.js

9.All Applications :

Placement Portal

DashboardCompaniesJobsApplicationsLogout

All Applications

#	Student	Email	Job Role	Company	Applied On	Current Status	Update Status
1	Pradeep Kumar	pradeepKumar081@gmail.com	Support	Software Tech	01 May 2026	SHORTLISTED	Shortlisted Update
2	Pradeep Taware	pradeeptaware6269@gmail.com	software Developer	TCS	01 May 2026	APPLIED	Applied Update
3	Pradeep Kumar	pradeepKumar081@gmail.com	Software Tester	Cognizant	01 May 2026	SHORTLISTED	Shortlisted Update
4	Aarav Patil	student001@viitplacement.com	Support	Software Tech	01 May 2026	REJECTED	Rejected Update
5	Vivaan Sharma	student002@viitplacement.com	Software Engineer	TCS	01 May 2026	SELECTED	Selected Update
6	Aditya Jadhav	student003@viitplacement.com	Associate Engineer	Software Tech	01 May 2026	APPLIED	Applied Update
7	Vihaan Kulkarni	student004@viitplacement.com	Full Stack Developer	Cognizant	01 May 2026	SHORTLISTED	Shortlisted Update

4. Database Tables :

1.Admin Table :

id	full_name	password	username
1	System Admin	Pass@123	pravinbagal123
NULL	NULL	NULL	NULL

2. Student Table :

id	address	branch	cgpa	course	email	full_name	passing_year	password
1	At Post Pune 411007	CSE	9.5	B.Tech	pradeeptaware6269@gmail.com	Pradeep Taware	2024	Pass@123
2	Baramati 413115	IT	8.5	MCA	pradeepKumar081@gmail.com	Pradeep Kumar	2024	Pass@123
3	Student Address 1, Pune, Maharashtra	Computer Science	6.14	BCA	student001@viitplacement.com	Aarav Patil	2024	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
4	Student Address 2, Pune, Maharashtra	Information Technology	9.31	BBA	student002@viitplacement.com	Vivaan Sharma	2025	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
5	Student Address 3, Pune, Maharashtra	AI & DS	6.11	BSc CS	student003@viitplacement.com	Aditya Jadhav	2026	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
6	Student Address 4, Pune, Maharashtra	Electronics	8.61	MCA	student004@viitplacement.com	Vihaan Kulkarni	2027	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
7	Student Address 5, Pune, Maharashtra	Mechanical	6.75	BCA	student005@viitplacement.com	Arjun Pawar	2024	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
8	Student Address 6, Pune, Maharashtra	Computer Science	9.9	BBA	student006@viitplacement.com	Sai Deshmukh	2025	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
9	Student Address 7, Pune, Maharashtra	Information Technology	7.24	BSc CS	student007@viitplacement.com	Krishna Joshi	2026	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
10	Student Address 8, Pune, Maharashtra	AI & DS	8.52	MCA	student008@viitplacement.com	Rohan Shinde	2027	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
11	Student Address 9, Pune, Maharashtra	Electronics	6.88	BCA	student009@viitplacement.com	Yash More	2024	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;
12	Student Address 10, Pune, Maharashtra	Mechanical	6.83	BBA	student010@viitplacement.com	Om Chavan	2025	\$2a\$10\$7QJ8mD3nKxQ9sVfL2eTn;

3.Job Table :

id	application_deadline	job_description	min_cgpa	package_offered	passing_year	required_branch	required_skills	role
1	2026-05-08	Good command on the java and oops concepts	6.5	4.5	2025	IT	Java	software D
2	2026-06-10		7	3.5	2024	IT	Selenium	Software Tr
3	2026-05-08		6	6	2024	IT	SQL,Java	Support
4	2026-05-20	Entry-level software development role focused ...	6.5	4.5	2025	Computer Science	Java, Spring Boot, MySQL	Software E
5	2026-05-22	Graduate engineer role for enterprise applicatio...	6	4.2	2025	Information Technology	Java, SQL, OOP	Associate E
6	2026-05-24	Work on modern web applications and scalable ...	6.2	4.8	2025	Computer Science	JavaScript, React, Node.js	Full Stack E
7	2026-05-26	Support cloud transformation projects and build...	6.5	5.2	2025	Information Technology	Java, Spring Boot, REST API	Cloud Assc
8	2026-05-28	Develop and maintain enterprise software soluti...	6.3	4.6	2025	Computer Science	Java, SQL, Hibernate	Graduate E
9	2026-05-30	Work on digital transformation and client-facing ...	6.4	4.7	2025	AI & DS	Python, SQL, Data Analysis	Data Analy
10	2026-06-02	Develop scalable applications and support enter...	6	4.3	2025	Computer Science	Java, DBMS, OOP	Programme
11	2026-06-04	Support software engineering and automation f...	6.2	4.9	2025	Information Technology	Python, Automation, SQL	Automator
12	2026-06-06	Build and maintain cloud and enterprise softwar...	6.5	5.8	2025	Computer Science	Cloud, Java, Linux	Cloud Engi

4.Companies Table :

id	description	industry	location	name	website
1	For Fresher students only...	IT/Software	Pune	TCS	https://www.tcs.com/careers/india/tcs-all-india-...
2		Banking/Finance	Baramati	Software Tech	https://www.Softwaretech.com/careers
3	open for freshers only	E-commerce	Chennai	Cognizant	https://www.cognizant.com/careers
4	Global IT services, consulting and business solut...	Information Technology	Mumbai, Maharashtra	Tata Consultancy Services	https://www.tcs.com
5	Digital services and consulting company deliveri...	Information Technology	Bengaluru, Karnataka	Infosys	https://www.infosys.com
6	Leading technology services and consulting com...	Information Technology	Bengaluru, Karnataka	Wipro	https://www.wipro.com
7	Global professional services company in digital, ...	Information Technology	Pune, Maharashtra	Accenture	https://www.accenture.com
8	Technology transformation and consulting servi...	Information Technology	Mumbai, Maharashtra	Capgemini	https://www.capgemini.com
9	IT services and consulting company focused on ...	Information Technology	Chennai, Tamil Nadu	Cognizant	https://www.cognizant.com
10	IT and business process outsourcing services co...	Information Technology	Pune, Maharashtra	Tech Mahindra	https://www.techmahindra.com
11	Global technology company delivering software ...	Information Technology	Noida, Uttar Pradesh	HCLTech	https://www.hcltech.com
12	Enterprise technology and consulting services p...	Information Technology	Bengaluru, Karnataka	IBM India	https://www.ibm.com/in-en
13	Cloud infrastructure and enterprise software co...	Information Technology	Bengaluru, Karnataka	Oracle India	https://www.oracle.com/in

5.Application Table :

id	applied_at	status	job_id	student_id
1	2026-05-01 09:10:45.484317	SHORTLISTED	3	2
2	2026-05-01 09:15:00.000000	REJECTED	1	1
3	2026-05-01 09:45:00.000000	SHORTLISTED	2	2
4	2026-05-01 10:10:00.000000	REJECTED	3	3
5	2026-05-01 10:35:00.000000	SELECTED	4	4
6	2026-05-01 11:00:00.000000	APPLIED	5	5
7	2026-05-01 11:20:00.000000	SHORTLISTED	6	6
8	2026-05-01 11:45:00.000000	APPLIED	7	7
9	2026-05-01 12:10:00.000000	REJECTED	8	8
10	2026-05-01 12:35:00.000000	SELECTED	9	9
11	2026-05-01 01:00:00.000000	APPLIED	10	10
12	2026-05-01 01:20:00.000000	SHORTLISTED	11	11
13	2026-05-01 01:45:00.000000	REJECTED	12	12

5.Conclusion :

The Placement Portal project is a complete web-based solution designed to simplify and automate the campus recruitment process for students, administrators, and companies. The system provides an efficient platform where students can register, upload their academic details, and apply for job opportunities, while administrators can manage student records, company information, and placement activities in a centralized manner.

This project reduces manual work, saves time, improves data accuracy, and enhances communication between students and recruiters. By using technologies such as Java, Spring Boot, MySQL, HTML, CSS, and REST APIs, the portal ensures secure data management and smooth system performance.

The Placement Portal also helps educational institutions maintain proper placement records, generate reports, and monitor student recruitment activities effectively. Overall, the project demonstrates the practical implementation of full-stack web development concepts and provides a scalable foundation for future enhancements such as email notifications, resume analysis, interview scheduling, and AI-based job recommendations.

6. Advantages :

1. **Centralized Management**

All student, company, and placement information is stored in one system, making management easier and more organized.

2. **Time Saving**

The portal automates manual tasks such as student registration, job application processing, and placement record management.

3. **Easy Communication**

Students and administrators can receive placement updates, job notifications, and company details quickly.

4. **Paperless Process**

Reduces paperwork by maintaining digital records of students, resumes, and placement activities.

5. **Improved Accuracy**

Minimizes human errors in maintaining placement data and student records.

6. **Quick Access to Information**

Students can easily view job opportunities, eligibility criteria, and application status anytime.

7. **Secure Data Storage**

Student and company data are stored securely in the database with proper authentication.

8. **Efficient Recruitment Process**

Companies can shortlist eligible candidates faster using filters such as percentage, skills, and qualifications.

9. **Report Generation**

Administrators can generate placement reports and analyze placement statistics easily.

10. **Scalable System**

The portal can be upgraded with additional features such as email notifications, interview scheduling, and AI-based recommendations in the future.

7. Future Enhancements :

1. **Email and SMS Notifications**

Add automatic email and SMS alerts for job postings, interview schedules, and placement updates.

2. **Resume Upload and Verification**

Allow students to upload resumes and enable admin verification for better recruitment management.

3. **Online Aptitude Test Module**

Integrate online tests and coding assessments for company recruitment processes.

4. **Interview Scheduling System**

Provide automatic interview scheduling with date, time, and meeting link notifications.

5. **AI-Based Job Recommendation**

Use Artificial Intelligence to recommend suitable jobs based on student skills, marks, and interests.

6. **Company Dashboard**

Create a separate dashboard for companies to post jobs, shortlist candidates, and track applications.

7. **Student Performance Analytics**

Add graphical reports and analytics to track student placement performance and statistics.

8. **Mobile Application Support**

Develop Android and iOS mobile applications for easy access to the portal.

9. **Cloud Database Integration**

Store data on cloud platforms for better scalability, backup, and security.

10. **Document Verification System**

Add digital verification for certificates, mark sheets, and other student documents.

11. **Chat Support System**

Integrate chatbot or live chat support for instant student assistance.

12. **Multi-College Support**

Extend the portal to support multiple colleges and institutions within a single system.

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